

AGENTS.md

Repository structure

```
/
├── backend/crminsight/    # Spring Boot 4.0.6 + Java 21 (Maven)
│   ├── pom.xml
│   └── src/main/java/com/uern/tep/crminsight/
│       ├── CrminsightApplication.java    # entrypoint
│       ├── config/
│       │   ├── SecurityConfig.java        # JWT stateless, CORS, rotas
│       │   ├── JwtAuthenticationFilter.java
│       │   ├── PasswordEncoderConfig.java
│       │   └── DataInitializer.java        # seed admin:admin123
│       ├── controller/
│       │   ├── AuthController.java        # POST /api/auth/login
│       │   ├── ClienteController.java
│       │   ├── VendedorController.java
│       │   ├── VendaController.java
│       │   ├── InteracaoController.java
│       │   ├── AnaliticoController.java
│       │   └── MeuAnaliticoController.java # GET /api/analitico/meu/*
│       ├── UsuarioController.java        # CRUD /api/usuarios (ADMIN)
│       ├── model/
│       │   ├── entity/
│       │   │   ├── Cliente.java
│       │   │   ├── Vendedor.java
│       │   │   ├── Venda.java
│       │   │   ├── Interacao.java
│       │   │   └── Usuario.java          # JWT auth
│       │   ├── dto/request/ (records com @Valid)
│       │   ├── dto/response/ (records)
│       │   └── enums/ (RankVendedor, StatusVenda, CanalInteracao,
│       │       RoleUsuario)
│       ├── handler/
│       │   └── GlobalExceptionHandler.java # JSON padronizado pra
│       │       erros
│       ├── repository/ (Spring Data JPA)
│       └── service/
│           ├── ClienteService.java
│           ├── VendedorService.java
│           ├── VendaService.java
│           ├── InteracaoService.java
│           ├── ScoreService.java
│           ├── AnaliticoService.java
│           ├── UsuarioService.java
│           └── JwtService.java
```

```

└─ frontend/crminsight/    # React 19 + TypeScript 6 + Vite 8
  └─ package.json
  └─ vite.config.ts
  └─ src/
    └─ main.tsx            # SPA entrypoint
    └─ App.tsx

```

Key commands

Frontend (**frontend/crminsight/**)

Command	Action
<code>npm run dev</code>	Start Vite dev server (HMR)
<code>npm run build</code>	<code>tsc -b && vite build</code> (typecheck first, then bundle)
<code>npm run lint</code>	<code>eslint .</code> (flat config; eslint 10, typescript-eslint 8)
<code>npm run preview</code>	Preview production build

Backend (**backend/crminsight/**)

```

./mvnw spring-boot:run      # dev server
./mvnw test                 # run all tests
./mvnw test -Dtest=TestClassName # single test class
./mvnw clean install        # full build (skipITs if integration tests
exist)

```

Framework quirks

- **Frontend build fails if `tsc` errors** — `npm run build` runs `tsc -b` before `vite build`. Always run `tsc --noEmit` to check types first.
- **TypeScript 6** — uses `erasableSyntaxOnly` (no `enum`, no `namespace`); `verbatimModuleSyntax` (must use `import type`).
- **ESLint flat config** — `eslint.config.js` at root; no `.eslintrc`. Lint scope is entire project excluding `dist/`.
- **Backend uses Spring Boot 4 + Java 21** — Lombok annotation processing is configured via `maven-compiler-plugin`. No `application.yml` yet, only `application.properties`.
- **Dual DB** — H2 (runtime scope) and PostgreSQL (runtime scope) are both declared. Either can be activated via Spring profile or classpath. No profile config exists yet.
- **Maven wrapper** — use `./mvnw`, not system `mvn`.
- **No monorepo tool** — frontend and backend are independent projects sharing a root directory. No workspace config linking them.

`.vscode/settings.json`

Enables `java.compile.nullAnalysis.mode: "automatic"` — relevant if editing Java nullability annotations.

What is NOT in this repo

- No Docker / docker-compose / infra config
- No CI workflows
- No integration test fixtures or service dependencies
- No database migration tooling (JPA auto-DDL assumed)
- No git history yet (no `.git/` directory)